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Course/Code: DSA0604 - DATA HANDLING AND VISUALIZATION

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SET 11: PRODUCT CATEGORY ANALYSIS

1) Funnel-style Chart – Sales by Category

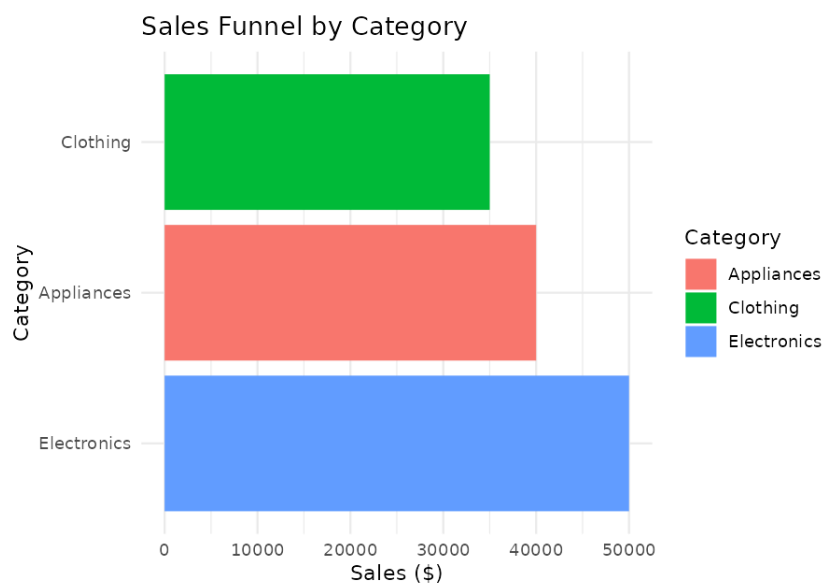
CODE:

```
library(ggplot2)

category <- data.frame(
  Category = c("Electronics", "Clothing", "Appliances"),
  Sales = c(50000, 35000, 40000)
)

ggplot(category, aes(x = reorder(Category, -Sales), y = Sales, fill = Category)) +
  geom_col() +
  coord_flip() +
  labs(title = "Sales Funnel by Category", x = "Category", y = "Sales ($)") +
  theme_minimal()
```

OUTPUT:



2) Pie Chart – Sales Distribution by Category

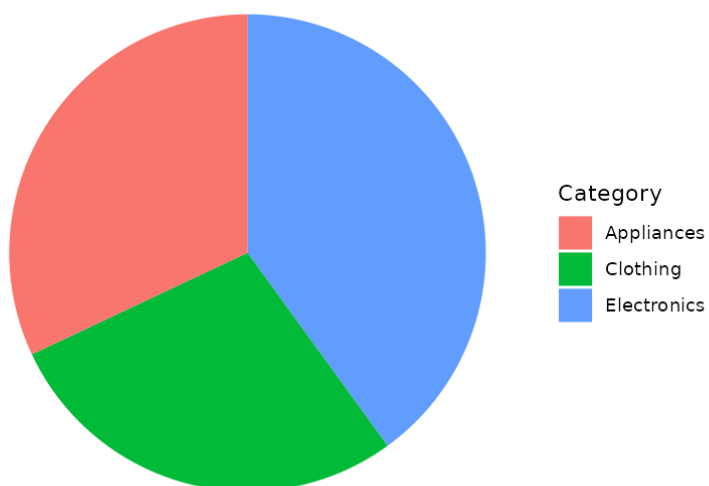
CODE:

```
library(ggplot2)

ggplot(category, aes(x = "", y = Sales, fill = Category)) +
  geom_bar(stat = "identity", width = 1) +
  coord_polar("y") +
  labs(title = "Sales Distribution by Category") +
  theme_void()
```

OUTPUT:

Sales Distribution by Category



3) Table – Sales Data per Category

CODE:

```
print(category)
```

OUTPUT:

	Category	Sales
1	Electronics	50000
2	Clothing	35000
3	Appliances	40000

4) Dashboard – Funnel + Pie + Table

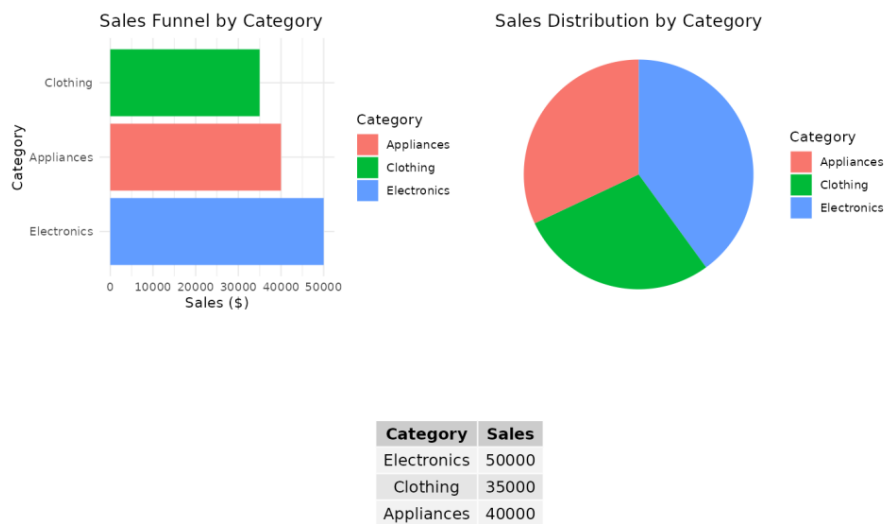
CODE:

```
ggplot(category, aes(x = reorder(Category, -Sales), y = Sales, fill = Category)) +  
  geom_col() + coord_flip() +  
  labs(title = "Sales Funnel by Category")
```

```
ggplot(category, aes(x = "", y = Sales, fill = Category)) +  
  geom_bar(stat = "identity", width = 1) + coord_polar("y") +  
  labs(title = "Sales Distribution by Category")
```

```
print(category)
```

OUTPUT:



SET 12: WEBSITE TRAFFIC ANALYSIS

1) Line Chart – Daily Page Views Trend

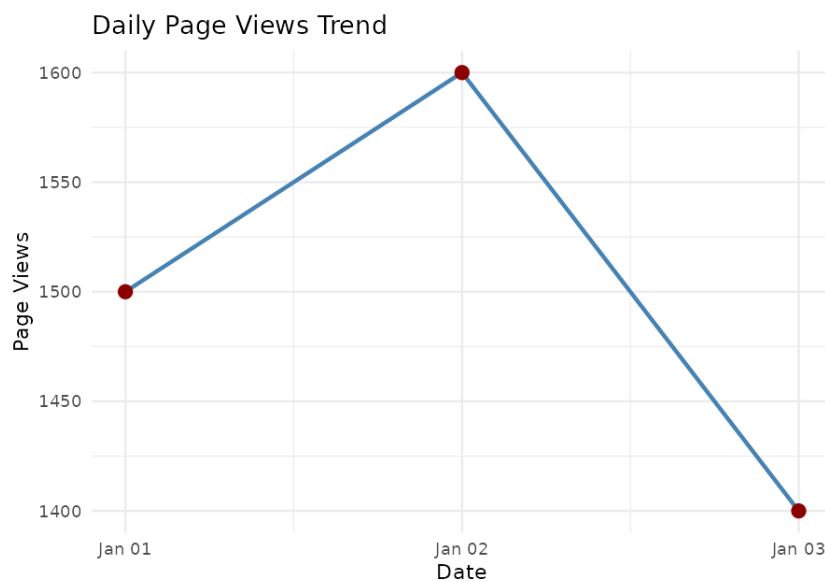
CODE:

```
library(ggplot2)

web12 <- data.frame(
  Date = as.Date(c("2023-01-01", "2023-01-02", "2023-01-03")),
  PageViews = c(1500, 1600, 1400),
  CTR = c(2.3, 2.7, 2.0)
)

ggplot(web12, aes(x = Date, y = PageViews)) +
  geom_line(color = "steelblue", linewidth = 1) +
  geom_point(color = "darkred", size = 3) +
  labs(title = "Daily Page Views Trend", x = "Date", y = "Page Views") +
  theme_minimal()
```

OUTPUT:



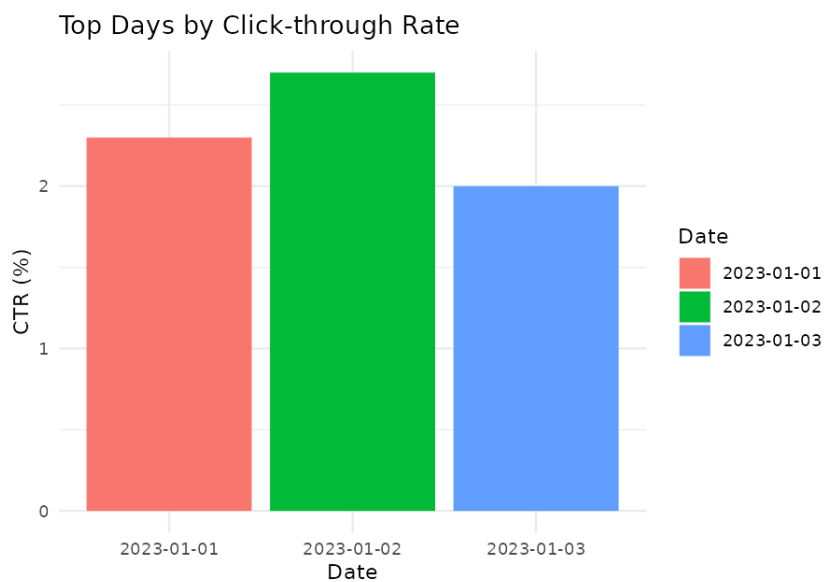
2) Bar Chart – Top Days by Click-through Rate

CODE:

```
library(ggplot2)

ggplot(web12, aes(x = as.character(Date), y = CTR, fill = as.character(Date))) +
  geom_bar(stat = "identity") +
  labs(title = "Top Days by Click-through Rate", x = "Date", y = "CTR (%)", fill = "Date") +
  theme_minimal()
```

OUTPUT:



3) Table – Website Traffic Data

CODE:

```
print(web12)
```

OUTPUT:

```
> print(web12)
```

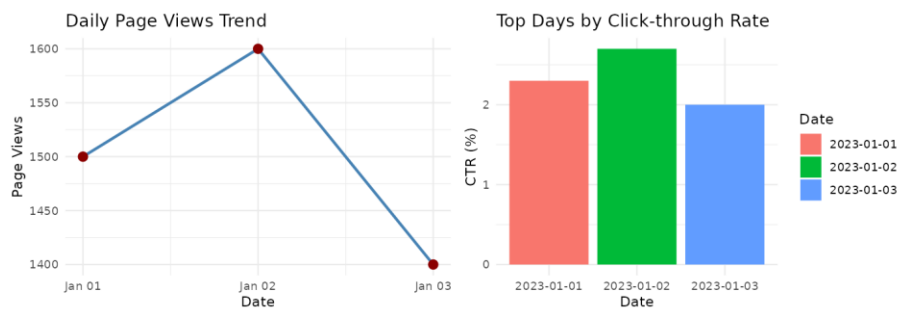
	Date	PageViews	CTR
1	2023-01-01	1500	2.3
2	2023-01-02	1600	2.7
3	2023-01-03	1400	2.0

4) Dashboard – Line + Bar + Table

CODE:

```
ggplot(web12, aes(x = Date, y = PageViews)) +  
  geom_line(color = "steelblue") + geom_point(color = "darkred", size = 3) +  
  labs(title = "Daily Page Views Trend")  
  
ggplot(web12, aes(x = as.character(Date), y = CTR, fill = as.character(Date))) +  
  geom_bar(stat = "identity") +  
  labs(title = "Top Days by Click-through Rate")  
  
print(web12)
```

OUTPUT:



Date	PageViews	CTR
2023-01-01	1500	2.3
2023-01-02	1600	2.7
2023-01-03	1400	2.0

SET 13: GEOGRAPHIC DATA ANALYSIS

1) Map Chart – City Population Distribution

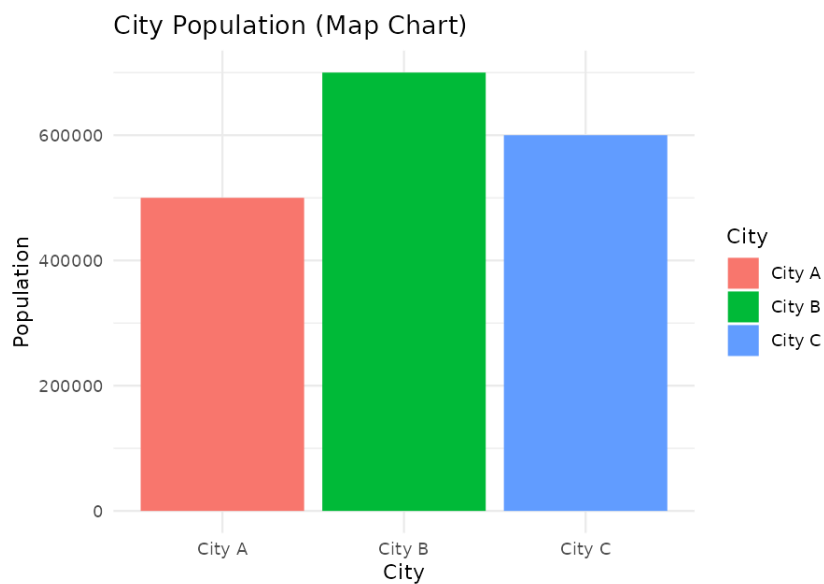
CODE:

```
library(ggplot2)
```

```
geo <- data.frame(  
  City = c("City A", "City B", "City C"),  
  Population = c(500000, 700000, 600000),  
  AvgTemp = c(75, 68, 80),  
  Elevation = c(1000, 800, 1200)  
)
```

```
ggplot(geo, aes(x = City, y = Population, fill = City)) +  
  geom_col() +  
  labs(title = "City Population (Map Chart)", x = "City", y = "Population") +  
  theme_minimal()
```

OUTPUT:



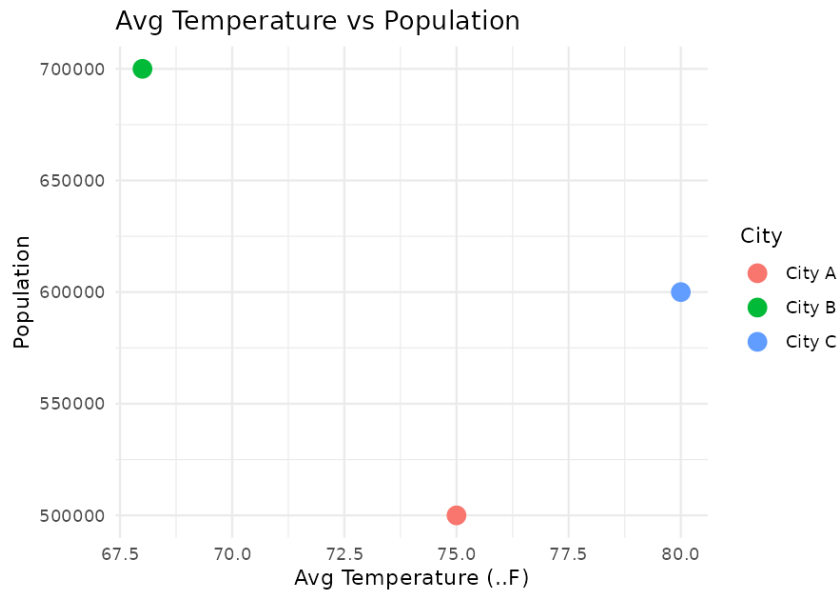
2) Scatter Plot – Avg Temperature vs Population

CODE:

```
library(ggplot2)
```

```
ggplot(geo, aes(x = AvgTemp, y = Population, color = City)) +  
  geom_point(size = 4) +  
  labs(title = "Avg Temperature vs Population", x = "Avg Temperature (°F)", y = "Population") +  
  theme_minimal()
```

OUTPUT:



3) Table – Geographic Data per City

CODE:

```
print(geo)
```

OUTPUT:

```
> print(geo)
```

City	Population	AvgTemp	Elevation
City A	500000	75	1000
City B	700000	68	800
City C	600000	80	1200

4) Dashboard – Map Chart + Scatter + Table

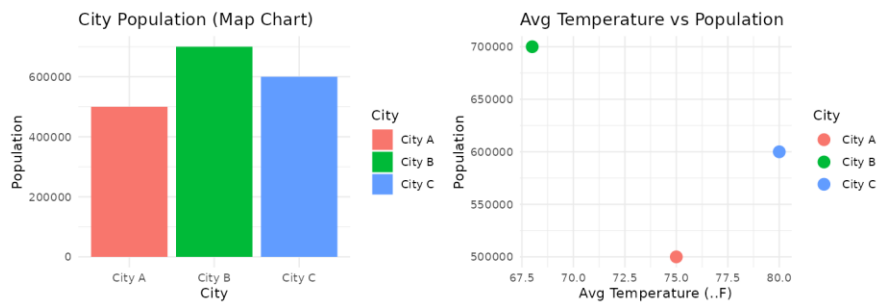
CODE:

```
ggplot(geo, aes(x = City, y = Population, fill = City)) +  
  geom_col() + labs(title = "City Population (Map Chart)")
```

```
ggplot(geo, aes(x = AvgTemp, y = Population, color = City)) +  
  geom_point(size = 4) + labs(title = "Avg Temperature vs Population")
```

```
print(geo)
```

OUTPUT:



City	Population	AvgTemp	Elevation
City A	500000	75	1000
City B	700000	68	800
City C	600000	80	1200

SET 14: SURVEY RESULTS ANALYSIS

1) Stacked Bar Chart – Distribution of Answers for Question 1

CODE:

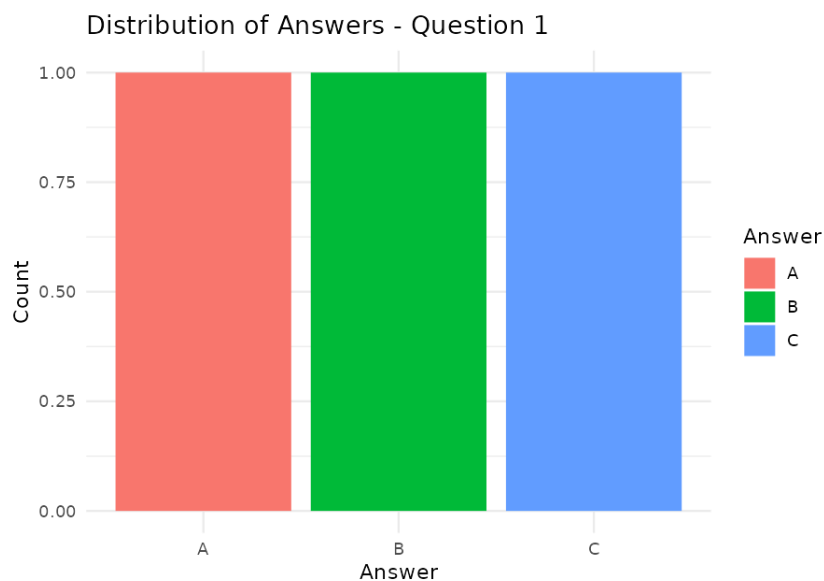
```
library(ggplot2)
```

```
survey14 <- data.frame(  
  Respondent = 1:3,  
  Question1 = c("A","B","C"),  
  Question2 = c("B","A","A"),  
  Question3 = c("C","D","B")  
)
```

```
q1_counts <- as.data.frame(table(survey14$Question1))  
colnames(q1_counts) <- c("Answer","Count")
```

```
ggplot(q1_counts, aes(x = Answer, y = Count, fill = Answer)) +  
  geom_bar(stat = "identity") +  
  labs(title = "Distribution of Answers - Question 1", x = "Answer", y = "Count") +  
  theme_minimal()
```

OUTPUT:



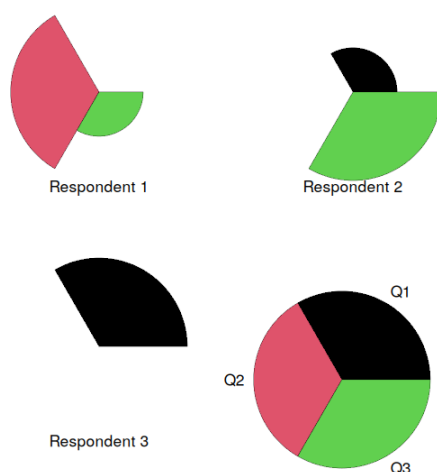
2) Radar Chart – Overall Pattern of Responses

CODE:

```
resp_matrix <- data.frame(  
  Q1 = as.numeric(factor(survey14$Question1, levels = c("A","B","C","D"))),  
  Q2 = as.numeric(factor(survey14$Question2, levels = c("A","B","C","D"))),  
  Q3 = as.numeric(factor(survey14$Question3, levels = c("A","B","C","D")))  
)  
rownames(resp_matrix) <- paste("Respondent", survey14$Respondent)  
  
stars(resp_matrix, key.loc = c(4.5, 2), main = "Radar Pattern of Survey Responses",  
  draw.segments = TRUE, len = 0.8)
```

OUTPUT:

Radar Pattern of Survey Responses



3) Table – Survey Response Data per Respondent

CODE:

```
print(survey14)
```

OUTPUT:

```
> print(survey14)
```

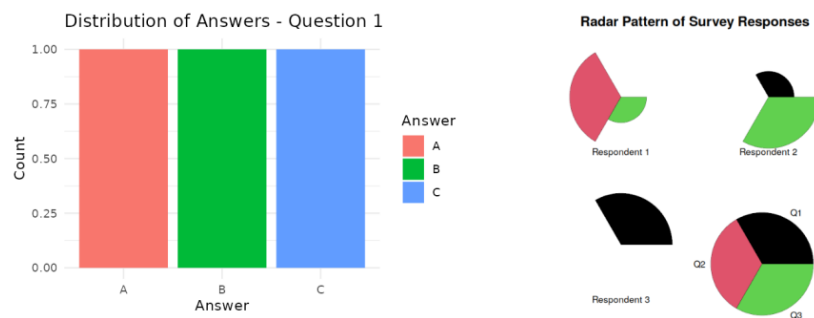
Respondent	Question1	Question2	Question3
1	A	B	C
2	B	A	D
3	C	A	B

4) Dashboard – Stacked Bar + Radar + Table

CODE:

```
ggplot(q1_counts, aes(x = Answer, y = Count, fill = Answer)) +  
  geom_bar(stat = "identity") + labs(title = "Question 1 Responses")  
  
stars(resp_matrix, key.loc = c(4.5, 2),  
      main = "Radar Pattern of Survey Responses", draw.segments = TRUE)  
  
print(survey14)
```

OUTPUT:



SET 15: PRODUCT SALES ANALYSIS

1) Grouped Bar Chart – Q1 Sales per Product

CODE:

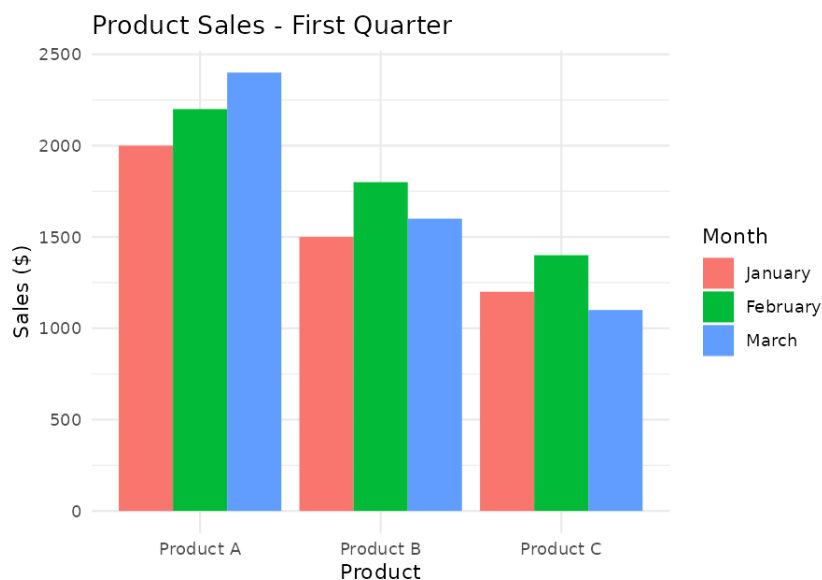
```
library(ggplot2)
library(reshape2)

sales15 <- data.frame(
  ProductID = 1:3,
  ProductName = c("Product A","Product B","Product C"),
  January = c(2000,1500,1200),
  February = c(2200,1800,1400),
  March = c(2400,1600,1100)
)

long15 <- melt(sales15, id.vars = c("ProductID","ProductName"),
  variable.name = "Month", value.name = "Sales")

ggplot(long15, aes(x = ProductName, y = Sales, fill = Month)) +
  geom_bar(stat = "identity", position = "dodge") +
  labs(title = "Product Sales - First Quarter", x = "Product", y = "Sales ($)") +
  theme_minimal()
```

OUTPUT:



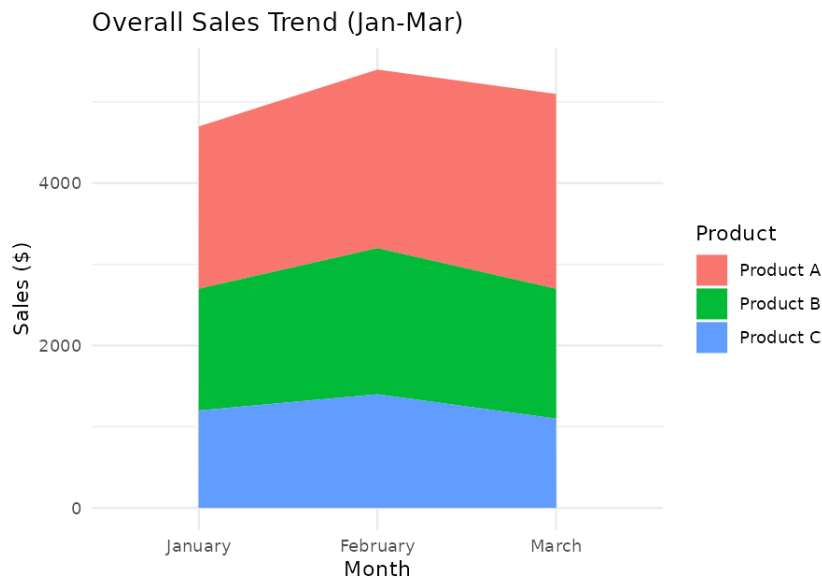
2) Stacked Area Chart – Overall Sales Trend

CODE:

```
library(ggplot2)
```

```
ggplot(long15, aes(x = Month, y = Sales, fill = ProductName, group = ProductName)) +  
  geom_area(position = "stack") +  
  labs(title = "Overall Sales Trend (Jan-Mar)", x = "Month", y = "Sales ($)") +  
  theme_minimal()
```

OUTPUT:



3) Table – Monthly Sales Data per Product

CODE:

```
print(sales15)
```

OUTPUT:

```
> print(sales15)
```

ProductID	ProductName	January	February	March
1	Product A	2000	2200	2400
2	Product B	1500	1800	1600
3	Product C	1200	1400	1100

4) Dashboard – Grouped Bar + Stacked Area + Table

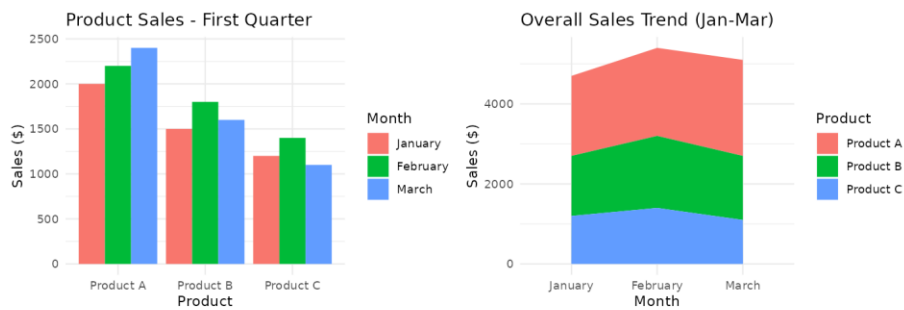
CODE:

```
ggplot(long15, aes(x = ProductName, y = Sales, fill = Month)) +  
  geom_bar(stat = "identity", position = "dodge") +  
  labs(title = "Monthly Product Sales")
```

```
ggplot(long15, aes(x = Month, y = Sales, fill = ProductName, group = ProductName)) +  
  geom_area() + labs(title = "Overall Sales Trend")
```

```
print(sales15)
```

OUTPUT:



ProductID	ProductName	January	February	March
1	Product A	2000	2200	2400
2	Product B	1500	1800	1600
3	Product C	1200	1400	1100

SET 16: CUSTOMER DEMOGRAPHICS ANALYSIS

1) Bar Chart – Distribution of Customer Ages

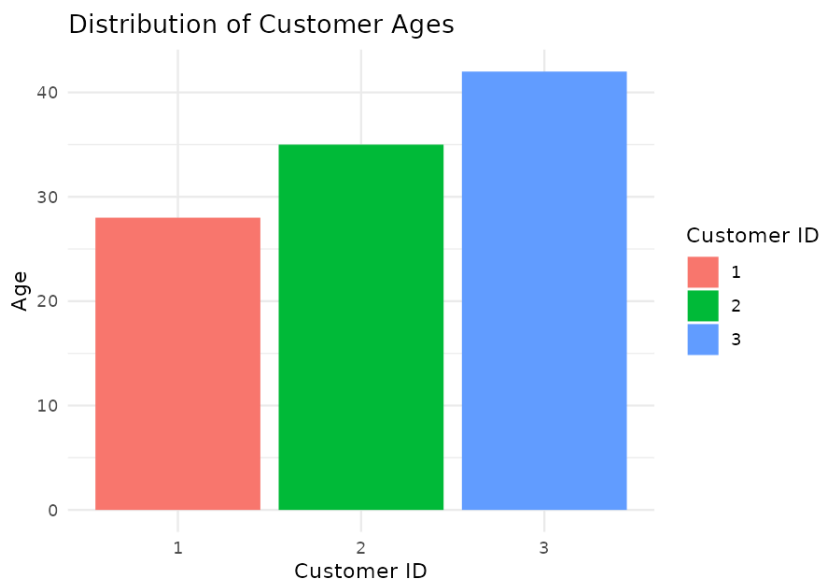
CODE:

```
library(ggplot2)
```

```
demo16 <- data.frame(  
  CustomerID = 1:3,  
  Age = c(28,35,42),  
  Gender = c("Female","Male","Female"),  
  Income = c(50000,60000,75000)  
)
```

```
ggplot(demo16, aes(x = factor(CustomerID), y = Age, fill = factor(CustomerID))) +  
  geom_bar(stat = "identity") +  
  labs(title = "Distribution of Customer Ages", x = "Customer ID", y = "Age", fill = "Customer ID") +  
  theme_minimal()
```

OUTPUT:



2) Pie Chart – Distribution by Gender

CODE:

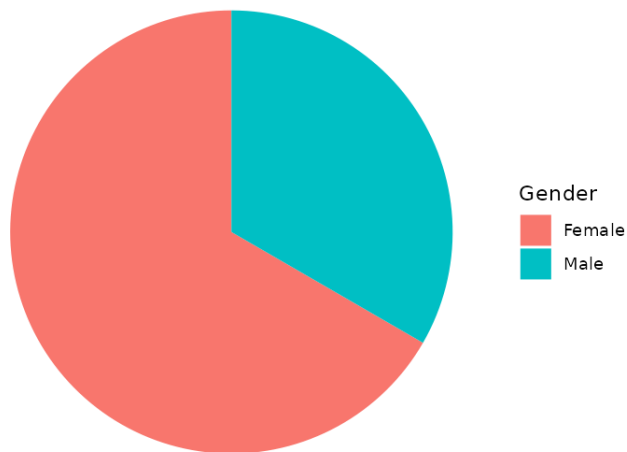
```
library(ggplot2)

gender_counts <- as.data.frame(table(demo16$Gender))
colnames(gender_counts) <- c("Gender","Count")

ggplot(gender_counts, aes(x = "", y = Count, fill = Gender)) +
  geom_bar(stat = "identity", width = 1) +
  coord_polar("y") +
  labs(title = "Customer Distribution by Gender") +
  theme_void()
```

OUTPUT:

Customer Distribution by Gender



3) Table – Demographic Information per Customer

CODE:

```
print(demo16)
```

OUTPUT:

```
> print(demo16)
```

CustomerID	Age	Gender	Income
1	28	Female	50000
2	35	Male	60000
3	42	Female	75000

4) Dashboard – Bar + Pie + Table

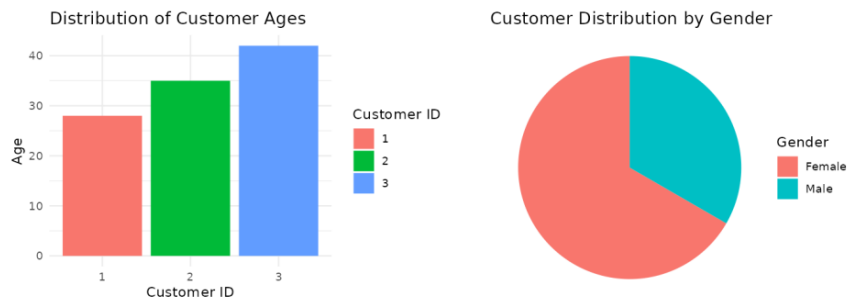
CODE:

```
ggplot(demo16, aes(x = factor(CustomerID), y = Age, fill = factor(CustomerID))) +  
  geom_bar(stat = "identity") + labs(title = "Customer Age Distribution")
```

```
gender <- table(demo16$Gender)  
pie(gender, labels = names(gender), main = "Gender Distribution")
```

```
print(demo16)
```

OUTPUT:



CustomerID	Age	Gender	Income
1	28	Female	50000
2	35	Male	60000
3	42	Female	75000

SET 17: EMPLOYEE PERFORMANCE ANALYSIS

1) Line Chart – Employee Performance Trend

CODE:

```
library(ggplot2)
```

```
emp17 <- data.frame(
```

```
  EmployeeID = 1:3,
```

```
  Department = c("Sales","HR","Marketing"),
```

```
  YearsOfService = c(5,3,7),
```

```
  PerformanceScore = c(85,92,78)
```

```
)
```

```
ggplot(emp17, aes(x = EmployeeID, y = PerformanceScore, group = 1)) +
```

```
  geom_line(color = "darkgreen", linewidth = 1) +
```

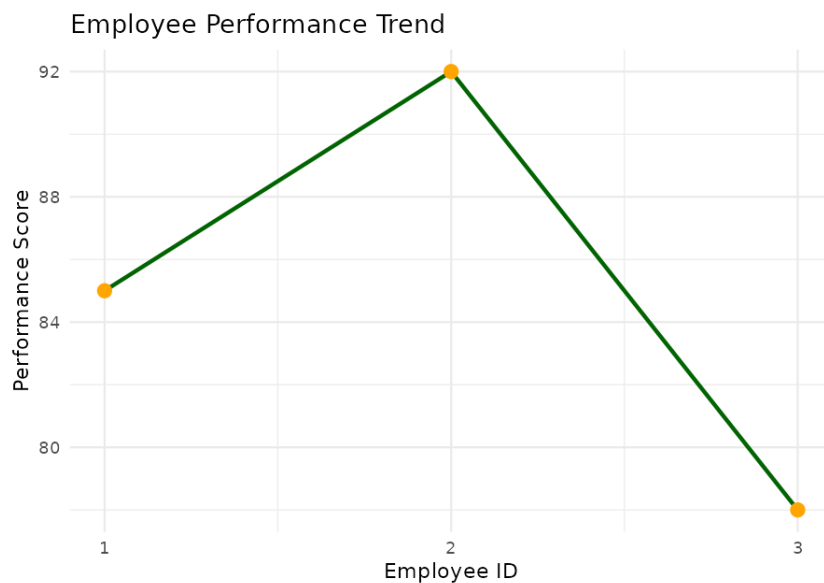
```
  geom_point(color = "orange", size = 3) +
```

```
  scale_x_continuous(breaks = emp17$EmployeeID) +
```

```
  labs(title = "Employee Performance Trend", x = "Employee ID", y = "Performance Score") +
```

```
  theme_minimal()
```

OUTPUT:



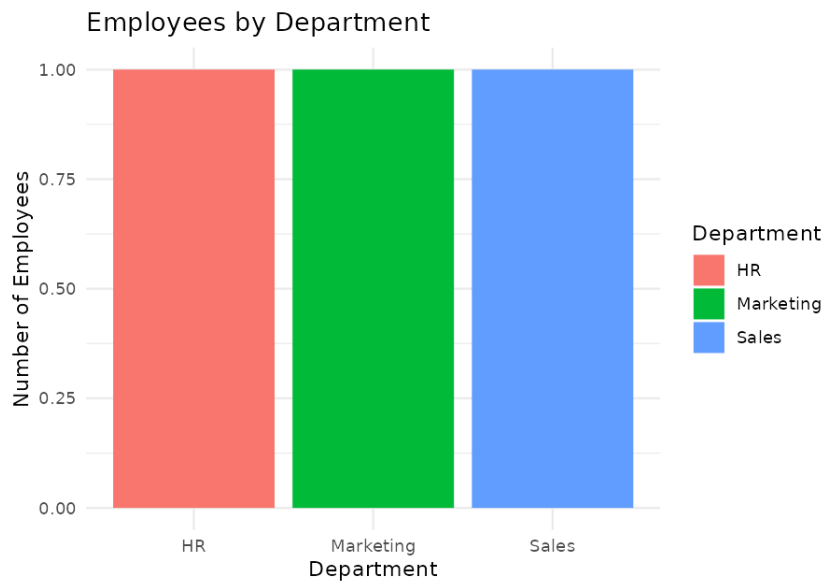
2) Bar Chart – Employees by Department

CODE:

```
library(ggplot2)
```

```
ggplot(emp17, aes(x = Department, fill = Department)) +  
  geom_bar() +  
  labs(title = "Employees by Department", x = "Department", y = "Number of Employees") +  
  theme_minimal()
```

OUTPUT:



3) Table – Performance Data per Employee

CODE:

```
print(emp17)
```

OUTPUT:

```
> print(emp17)
```

EmployeeID	Department	YearsOfService	PerformanceScore
1	Sales	5	85
2	HR	3	92
3	Marketing	7	78

4) Dashboard – Line + Bar + Table

CODE:

```
ggplot(emp17, aes(x = EmployeeID, y = PerformanceScore, group = 1)) +  
  geom_line(color = "darkgreen") + geom_point(color = "orange", size = 3) +  
  labs(title = "Employee Performance Trend")
```

```
ggplot(emp17, aes(x = Department, fill = Department)) +  
  geom_bar() + labs(title = "Employees by Department")
```

```
print(emp17)
```

OUTPUT:



EmployeeID	Department	YearsOfService	PerformanceScore
1	Sales	5	85
2	HR	3	92
3	Marketing	7	78

SET 18: PRODUCT INVENTORY MANAGEMENT

1) Bar Chart – Quantity of Each Product

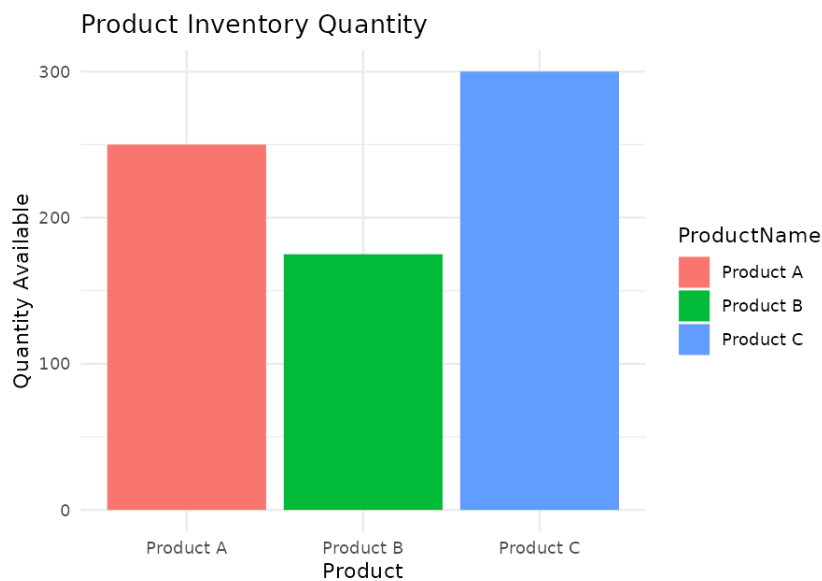
CODE:

```
library(ggplot2)
```

```
inv18 <- data.frame(  
  ProductID = 1:3,  
  ProductName = c("Product A", "Product B", "Product C"),  
  Category = c("Electronics", "Grocery", "Electronics"),  
  Quantity = c(250, 175, 300),  
  Price = c(20, 15, 18)  
)
```

```
ggplot(inv18, aes(x = ProductName, y = Quantity, fill = ProductName)) +  
  geom_bar(stat = "identity") +  
  labs(title = "Product Inventory Quantity", x = "Product", y = "Quantity Available") +  
  theme_minimal()
```

OUTPUT:



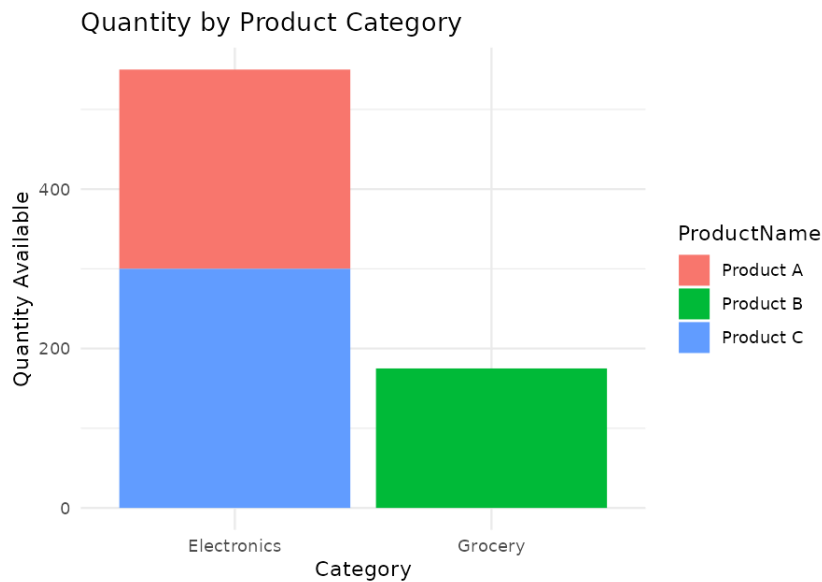
2) Stacked Bar Chart – Quantity by Product Category

CODE:

```
library(ggplot2)
```

```
ggplot(inv18, aes(x = Category, y = Quantity, fill = ProductName)) +  
  geom_bar(stat = "identity", position = "stack") +  
  labs(title = "Quantity by Product Category", x = "Category", y = "Quantity Available") +  
  theme_minimal()
```

OUTPUT:



3) Table – Inventory Data per Product

CODE:

```
print(inv18)
```

OUTPUT:

```
> print(inv18)
```

ProductID	ProductName	Category	Quantity	Price
1	Product A	Electronics	250	20
2	Product B	Grocery	175	15
3	Product C	Electronics	300	18

4) Dashboard – Bar + Stacked Bar + Table

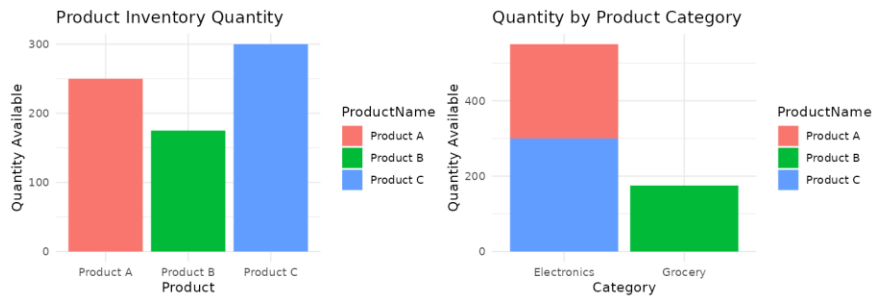
CODE:

```
ggplot(inv18, aes(x = ProductName, y = Quantity, fill = ProductName)) +  
  geom_bar(stat = "identity") + labs(title = "Inventory Quantity")
```

```
ggplot(inv18, aes(x = Category, y = Quantity, fill = ProductName)) +  
  geom_bar(stat = "identity", position = "stack") +  
  labs(title = "Quantity by Category")
```

```
print(inv18)
```

OUTPUT:



ProductID	ProductName	Category	Quantity	Price
1	Product A	Electronics	250	20
2	Product B	Grocery	175	15
3	Product C	Electronics	300	18

SET 19: SURVEY RESPONSES ANALYSIS

1) Grouped Bar Chart – Distribution of Answers for Question 1

CODE:

```
library(ggplot2)
library(reshape2)

survey19 <- data.frame(
  SurveyID = 1:3,
  Question1 = c("A","B","C"),
  Question2 = c("B","A","A"),
  Question3 = c("C","D","B")
)

q1_counts <- as.data.frame(table(survey19$Question1))
colnames(q1_counts) <- c("Answer","Count")

ggplot(q1_counts, aes(x = Answer, y = Count, fill = Answer)) +
  geom_bar(stat = "identity") +
  labs(title = "Distribution of Answers - Question 1", x = "Answer", y = "Count") +
  theme_minimal()
```

OUTPUT:



2) Stacked Bar Chart – Overall Distribution of Responses

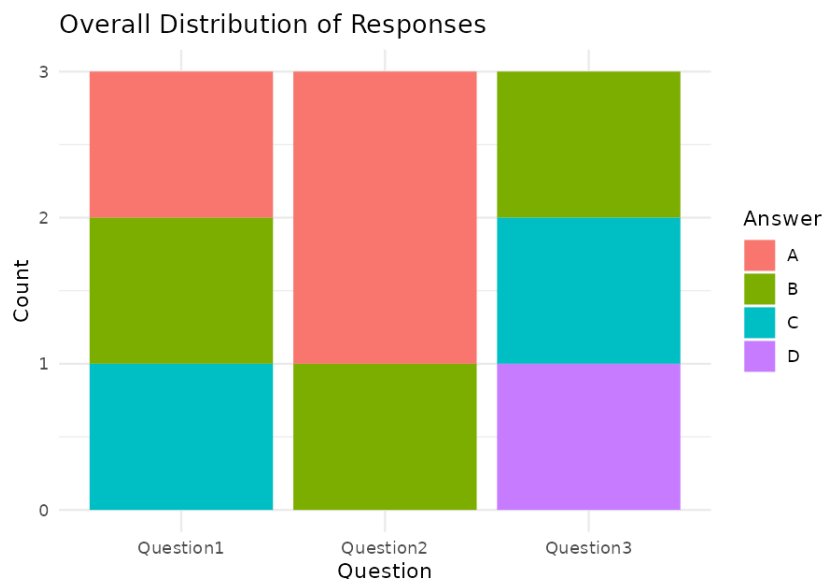
CODE:

```
library(ggplot2)

long_survey19 <- melt(survey19, id.vars = "SurveyID",
                      variable.name = "Question", value.name = "Answer")

ggplot(long_survey19, aes(x = Question, fill = Answer)) +
  geom_bar(position = "stack") +
  labs(title = "Overall Distribution of Responses", x = "Question", y = "Count") +
  theme_minimal()
```

OUTPUT:



3) Table – Survey Response Data per Survey

CODE:

```
print(survey19)
```

OUTPUT:

```
> print(survey19)
```

SurveyID	Question1	Question2	Question3
1	A	B	C
2	B	A	D
3	C	A	B

4) Dashboard – Grouped Bar + Stacked Bar + Table

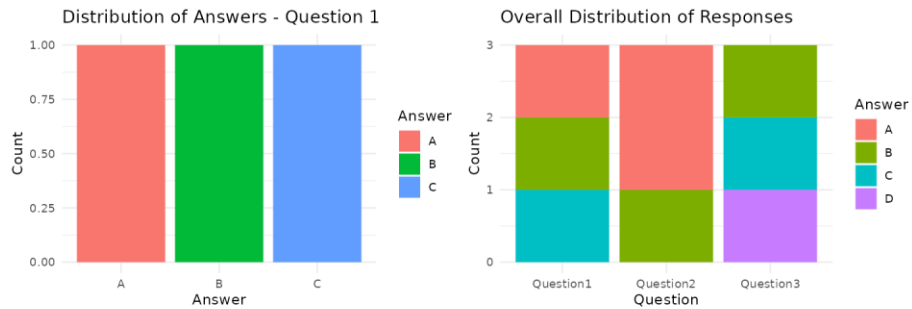
CODE:

```
ggplot(q1_counts, aes(x = Answer, y = Count, fill = Answer)) +  
  geom_bar(stat = "identity") + labs(title = "Question 1 Responses")
```

```
ggplot(long_survey19, aes(x = Question, fill = Answer)) +  
  geom_bar() + labs(title = "Overall Survey Responses")
```

```
print(survey19)
```

OUTPUT:



SurveyID	Question1	Question2	Question3
1	A	B	C
2	B	A	D
3	C	A	B

SET 20: STOCK ANALYSIS

1) Line Chart – Stock Prices Over Time

CODE:

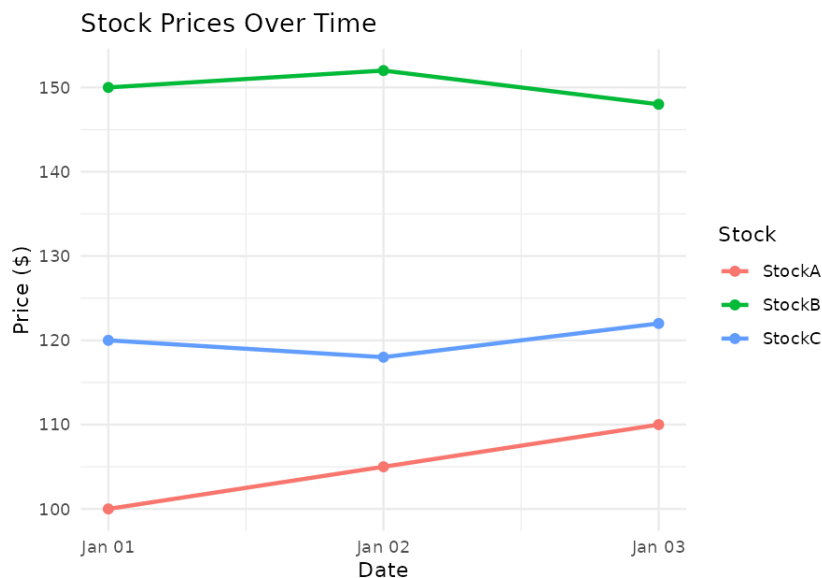
```
library(ggplot2)
library(reshape2)

stock <- data.frame(
  Date = as.Date(c("2023-01-01", "2023-01-02", "2023-01-03")),
  StockA = c(100, 105, 110),
  StockB = c(150, 152, 148),
  StockC = c(120, 118, 122)
)

long_stock <- melt(stock, id.vars = "Date", variable.name = "Stock", value.name = "Price")

ggplot(long_stock, aes(x = Date, y = Price, color = Stock, group = Stock)) +
  geom_line(linewidth = 1) +
  geom_point(size = 2) +
  labs(title = "Stock Prices Over Time", x = "Date", y = "Price ($)") +
  theme_minimal()
```

OUTPUT:



2) Bar Chart – Daily % Change in Stock A

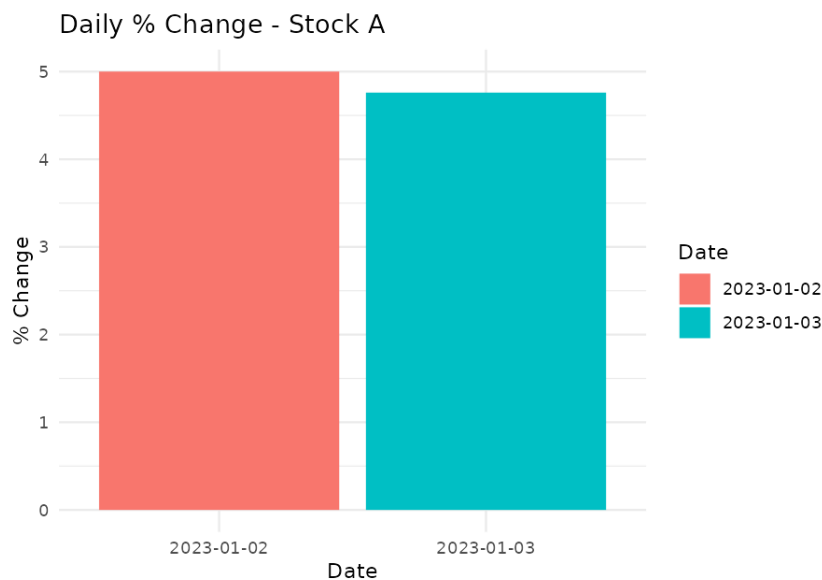
CODE:

```
library(ggplot2)
```

```
stock$PctChangeA <- c(NA, round(diff(stock$StockA) / head(stock$StockA, -1) * 100, 2))
```

```
ggplot(stock[-1, ], aes(x = as.character(Date), y = PctChangeA, fill = as.character(Date))) +  
  geom_bar(stat = "identity") +  
  labs(title = "Daily % Change - Stock A", x = "Date", y = "% Change", fill = "Date") +  
  theme_minimal()
```

OUTPUT:



3) Table – Stock Price Data

CODE:

```
print(stock[, c('Date','StockA','StockB','StockC')])
```

OUTPUT:

```
> print(stock[, c('Date','StockA','StockB','StockC')])
```

Date	StockA	StockB	StockC
2023-01-01	100	150	120
2023-01-02	105	152	118
2023-01-03	110	148	122

4) Dashboard – Line + Bar + Table

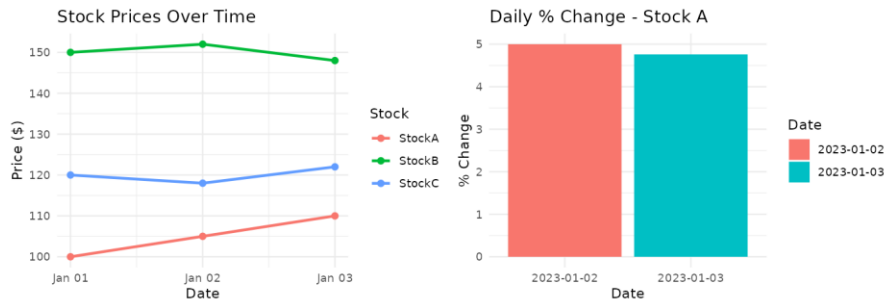
CODE:

```
ggplot(long_stock, aes(x = Date, y = Price, color = Stock, group = Stock)) +  
  geom_line() + geom_point() + labs(title = "Stock Prices Over Time")
```

```
ggplot(stock[-1, ], aes(x = as.character(Date), y = PctChangeA, fill = as.character(Date))) +  
  geom_bar(stat = "identity") + labs(title = "Daily % Change - Stock A")
```

```
print(stock)
```

OUTPUT:



Date	StockA	StockB	StockC
2023-01-01	100	150	120
2023-01-02	105	152	118
2023-01-03	110	148	122